

Step 9 One-Pager — Multi-Agent Reinforcement Learning

Research on the possibilities for applying Artificial Intelligence in computer games

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Problem. Steps 2–8 lived inside two-player zero-sum games, where a Nash strategy secures a value v^* against anyone and CFR provably converges to it. Step 9 is the pivot into the multi-agent world, where the defining difficulty is **non-stationarity**: every agent is learning at once, so from any one agent’s seat the environment (which contains the others) never holds still. This is the launch point for the thesis’s multi-agent contributions — dynamic opponent modeling (#1), safe exploitation without a minimax anchor (#2), and population-level evaluation (#3).

Approach. Build and compare the field’s four structural answers to non-stationarity on small, *exactly-solvable* testbeds, so every learner is graded against ground truth. **Independent learning** (the control that fails) on four canonical matrix games; **CTDE** — centralize the critic at training, decentralize the actor at execution — via MADDPG and MAPPO on cooperative tasks; **PSRO** — a game played over a *population* of policies (meta-Nash + best-response oracle), reusing Step 07’s exact best response as both oracle and exploitability metric — on Kuhn, Leduc, a matrix game, and a native Goofspiel; **learned communication** (CommNet); and **LOLA** (differentiate through the opponent’s learning step) on the Iterated Prisoner’s Dilemma. **All numbers below are measured.**

Key results (measured).

- *Independent learning fails exactly where theory says.* It converges to Nash on Prisoner’s Dilemma (defection), Stag Hunt (the risk-dominant corner), and Battle of the Sexes, but **Matching Pennies never converges** — its distance-to-Nash actually *grows* ($0.30 \rightarrow 0.48$), a sharper-than-predicted non-convergence (a contradicted prediction, §9 of the report).
- *PSRO is the game-theory $\leftrightarrow$ MARL bridge, and it works on the small games.* Exploitability $\rightarrow$ machine zero on **Kuhn** ($0.917 \rightarrow 2 \times 10^{-16}$, 6 rounds), $\rightarrow 0$ on the matrix game and Rock–Paper–Scissors. Self-play confirms the mechanism: on Kuhn the **average**-iterate NashConv falls $0.24 \rightarrow 0.031$ while the **last** iterate keeps oscillating — why a population/averaging is needed.
- *A centralized critic is a near-zero-variance teacher.* On CoopSignal the centralized critic’s residual is $3.2e-11$ vs the independent critic’s 0.077 — the CTDE variance-reduction claim, confirmed.
- *Communication clears the guessing ceiling.* CommNet with the channel ON scores 0.795 vs 0.204 OFF (ceiling $1/K = 0.2$) — a learned, not designed, protocol.
- *LOLA turns defectors into cooperators.* IPD per-step return rises from 1.04 (naive defection) to 2.82 (LOLA cooperation); zeroing the look-ahead recovers the naive gradient exactly (the mechanism check).
- *Honest negatives (kept predictions, §9).* PSRO on **Leduc** declines but hits a scaling wall ($4.75 \rightarrow 2.16$ after 20 rounds, not < 0.5); **Goofspiel K=4** oscillates rather than converging (a flagged code anomaly, documented not fixed); and on the **climbing game** no method reaches the optimum 11 (IL/MAPPO 7, MADDPG 5) — a centralized critic lowers variance but does not by itself solve hard-exploration coordination.

Thesis connection. PSRO is Step 2’s iterated best response lifted to a population and the empirical backbone here; LOLA is *dynamic* opponent modeling, the moving-target complement to Step 7’s static read (Contribution #1); PSRO’s meta-game is a population-level evaluation methodology (Contribution #3). The Leduc scaling wall echoes Step 8’s global-vs-local finding, and the place where every two-player guarantee stops — the **missing N>2 minimax anchor** — is exactly Contribution #2’s problem statement.

Open questions. Can an approximate (RL) oracle push Leduc PSRO close to Nash, and at what cost to the guarantee? What breaks the Goofspiel-K=4 oscillation (de-duplication? a mixed population? a different meta-solver?), and why does a centralized critic *hurt* on the climbing game? And underneath all of it: with no minimax value for $N>2$, what does “safe” mean for a population, and can PSRO’s meta-game supply the substitute (Contribution #2)?